

BOWIE, TX, USAWMO: **720622**Lat: **33.602N**Lon: **97.776W**Elev: **336**StdP: **97.36**Time Zone: **-6.00 (NAC)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.4	-13.9	1.2	-3.6	-11.2	1.5	1.4	11.1	6.8	10.2	8.2	2.7	0	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	37.9	21.8	37.0	22.1	35.3	22.9	25.6	32.0	25.1	31.8	24.6	31.3	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.7	28.3	23.3	18.8	27.7	22.8	18.2	27.4	80.8	31.9	78.4	31.9	76.2	31.4	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.4	7.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-11.5	26.5	2.9	1.0	-13.6	27.2	-15.3	27.8	-17.0	28.3	-19.1	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.2	8.6	13.6	17.5	21.4	26.5	28.2	28.6	24.6	17.7	12.9	7.6		
	DBStd	9.01	5.40	6.47	5.41	4.45	3.88	2.63	3.69	2.69	3.50	5.07	4.86	5.23		
	HDD10.0	412	141	95	26	4	0	0	1	0	0	7	26	111		
	HDD18.3	1487	376	277	163	66	17	0	7	0	3	73	171	335		
	CDD10.0	3267	23	55	138	229	352	496	564	576	438	244	114	36		
	CDD18.3	1302	1	4	17	41	111	246	312	317	191	52	8	2		
	CDH23.3	14742	8	43	149	379	1056	2761	3897	3902	1911	543	80	13		
	CDH26.7	7300	0	11	31	94	366	1353	2168	2192	896	171	17	1		
(14) Wind	WSAvg	3.1	3.2	3.6	3.9	4.0	3.4	3.0	2.4	2.3	2.4	2.9	3.3	3.2		
(15) Precipitation	PrecAvg	874	41	53	71	79	107	97	64	60	78	78	54	51		
	PrecMax	1172	91	169	193	222	227	269	162	151	170	192	183	184		
	PrecMin	552	0	0	0	10	16	27	5	0	4	0	4	0		
	PrecStd	177	26	47	47	52	54	61	47	43	49	55	51	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	27.8	29.8	32.1	34.0	37.5	40.6	41.1	37.4	33.3	28.9	25.2		
		MCWB	12.9	17.4	17.7	19.0	23.2	21.6	21.3	21.0	20.1	20.7	18.5	15.7		
	2%	DB	21.2	24.0	27.3	29.1	32.2	36.0	37.8	38.8	34.9	30.5	25.3	21.5		
		MCWB	12.2	15.0	16.7	18.8	21.9	22.2	21.7	21.6	22.2	20.3	16.4	14.0		
	5%	DB	18.0	21.5	24.9	27.3	30.3	34.4	36.9	37.2	33.0	28.4	23.0	19.0		
		MCWB	11.2	14.6	16.2	18.4	21.0	23.0	22.2	22.5	22.3	19.2	16.0	13.1		
	10%	DB	15.4	18.9	22.4	25.5	28.8	32.8	35.1	35.6	31.9	26.4	21.2	16.3		
		MCWB	10.0	13.5	15.0	17.6	21.1	23.2	23.3	23.0	21.7	18.2	15.1	12.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	19.1	20.2	21.9	25.0	26.5	26.4	25.9	25.3	23.4	21.6	18.3		
		MCDB	20.0	23.1	26.1	26.7	30.2	32.2	32.5	31.7	31.7	28.8	25.0	21.8		
	2%	WB	14.0	17.9	19.0	21.0	23.7	25.4	25.7	25.2	24.5	22.6	19.0	16.3		
		MCDB	17.7	21.7	24.3	26.3	28.9	31.4	32.2	32.6	30.6	27.7	22.0	19.5		
	5%	WB	12.2	16.4	17.8	20.1	22.8	24.8	25.2	24.7	23.9	21.6	17.9	14.6		
		MCDB	17.5	19.8	22.0	25.1	27.8	30.9	31.9	32.1	29.6	26.2	20.7	17.3		
	10%	WB	10.2	14.5	16.8	19.2	21.9	24.1	24.6	24.2	23.3	20.2	16.5	12.1		
		MCDB	14.9	18.3	20.8	23.7	26.8	30.2	31.4	31.6	28.8	24.3	20.4	15.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.1	12.5	12.7	11.2	11.4	12.4	12.4	11.7	12.8	12.8	11.1		
		MCDBR	16.8	15.5	14.7	14.7	13.6	13.1	14.4	14.3	13.6	14.8	14.6	15.1		
	5% WB	MCWBR	9.5	8.0	6.7	7.0	5.7	3.7	3.5	3.5	4.2	6.4	7.4	8.5		
		MCWBR	14.4	13.2	11.9	12.2	10.8	11.0	11.4	11.9	10.6	11.7	11.0	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.309	0.343	0.374	0.413	0.427	0.427	0.421	0.386	0.349	0.320	0.303		
	taud		2.514	2.495	2.419	2.358	2.313	2.311	2.347	2.334	2.390	2.477	2.496	2.499		
	Ebn at Noon		922	946	933	914	876	859	857	858	877	888	887	893		
	Edh at Noon		83	94	111	123	130	130	125	125	113	95	85	80		
(44) All-Sky Solar Radiation	RadAvg		2.92	3.50	4.60	5.59	6.01	6.73	6.79	6.23	5.18	4.03	3.10	2.54		
	RadStd		0.30	0.54	0.31	0.46	0.62	0.63	0.45	0.36	0.48	0.60	0.40	0.30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page